



SOP: Configuring i3Ai Setup for optimal accuracy

Objective

Establish a consistent process for configuring and verifying i3Ai detection settings on all cameras to ensure proper operation and detection capabilities at Northern Tool and Equipment's sites. This SOP outlines a clear process to ensure i3Ai cameras are configured correctly and maintain their detection capabilities. Regular checks and detailed documentation will help prevent future issues and enhance overall performance.

Scope

This SOP applies to all support staff responsible for configuring and maintaining i3Ai cameras across all Northern Tool and Equipment sites.

Limitations

Certain environmental and infrastructure events and conditions outside of i3 control may affect i3Ai accuracy and performance. This includes but is not limited to network performance, power fluctuations, camera lens/dome cleanliness, severe weather conditions, and other acts of God may interfere with i3Ai performance. i3 makes no claims to the accuracy and accepts no responsibility for the i3Ai performance under those circumstances.

Responsibilities

i3 Team	Global Axiom Team
i3 Technical Support: - Provide the Global Axiom team with clear i3Ai setup documentation. Note: Recommendations may change as i3Ai software evolves over time. <i>Due Date: June 20, 2024</i> i3 Technical Services Leads: - Provide the Global Axiom team with camera installation recommendations for best i3Ai accuracy. <i>Due Date: June 28, 2024</i> - Document issues identified on the NTE sites after a joint review with Global Axiom. Establish a clear timeline and responsibilities for any fixes that must be implemented. - Pass the video of a non-compliant event from the Global Axiom team to the i3's development team. Notify the Global Axiom team when the i3Ai algorithm/engine is re-trained and the update is available. i3 Executive Sponsor: - Approve the SOP and ensure compliance across all sites.	Global Technical Support: - Ensure recommended i3 software is installed on all NTE GPU-enabled servers. - Ensure clear camera view. - Ensure camera configurations are in compliance with i3's recommendations. - Verify i3Ai performance through audit and testing during the active alarm hours. Global Technical Services Leads: - Oversee the configuration process and conduct periodic audits. - Establish and maintain a regular camera cleaning schedule. - Review existing NTE sites with i3 Technical Services Lead for compliance. - Collect video from all non-compliant events (in *.i3d format), host on a shared SFTP location, and provide to i3's Team Leads as part of the escalation process. i3 will use non-compliant video clips to improve i3Ai algorithm performance and accuracy. - Install i3Ai software updates whenever new i3Ai algorithm/engine updates become available.
Both i3 and Global Axiom teams: NTE Site 658 Burleson will be physically audited. Detection accuracy, CMS connectivity, and successful signal alerts to the Global monitoring station will be confirmed during the audit. 4-5 additional sites will be upgraded following the same template by the end of the day on June 19, 2024	

Procedure

1. Infrastructure and Configuration Requirements:

○ Server Requirements:

1. i3Ai Server with GPU is required. GPU = GTX1050i (minimum), GTX1060 or better (recommended)

○ Software Requirements:

All current software packages can be downloaded from i3's Downloads website:

<https://i3international.com/download>

1. Install the latest SRX-Software (v7.3.2.120 as of June 19, 2024) on each server.
2. Install the NVIDIA driver (64-bit) on the i3Ai Server (GPU-enabled server).
3. Install the latest i3Ai Server software (v7.3.2.90 as of June 19, 2024)

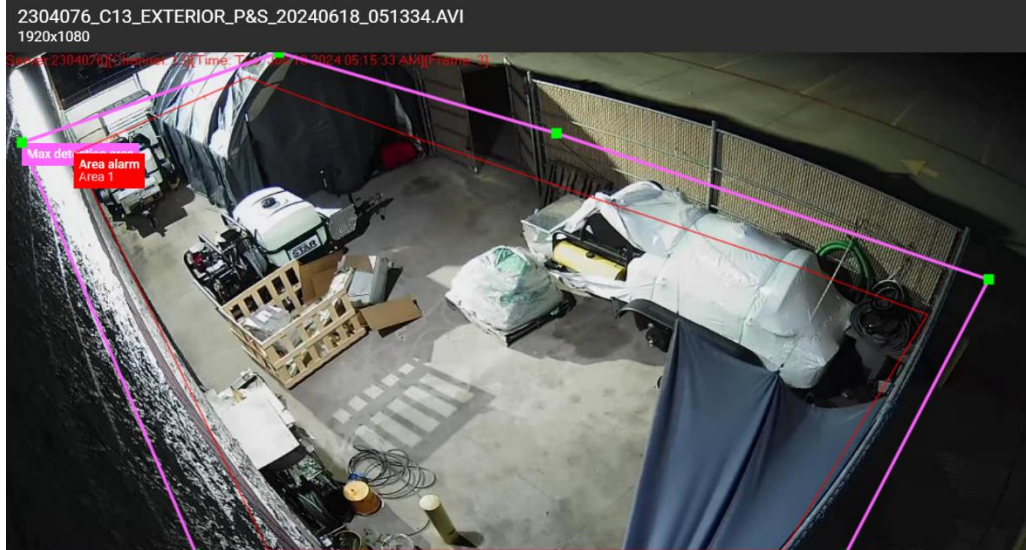
Note: All the latest i3 software packages have already been downloaded to 121 GPU-enabled NTE servers and can be found on the local D:\ partition, ready for installation.

○ i3Ai Setup Camera Configuration Requirements.

Camera installation recommendations, including best practices for camera angle and installation, as well as bitrate setting, maximum coverage area, etc. will be provided by i3 Team Leads.

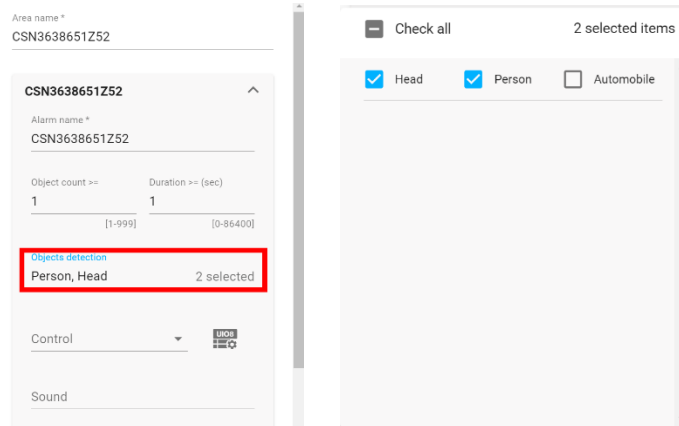
Due Date: **June 28, 2024**

1. Minimum acceptable resolution: 1280x960 (1.3MP).
2. Required i3 and Axis IP camera frame rate: 5 FPS.
3. Required HR16v2 encoder frame rate: 6 FPS.
4. "Max detection area" setting must be enabled.
5. Max detection area must be larger than the detection area.



6. Object detection must be set to "Head" and "Person".

7. Object count must be set to ≥ 1 , Duration must be set to ≥ 1



2. Verification Process

- **Post-Configuration Checks:**

1. Verify the frame rate settings for each camera.
2. Confirm the maximum detection settings are correctly applied.

- **Testing:**

1. Conduct an AI detection and alert test to ensure AI functionality during nighttime, wherever possible.
If a nighttime test is not possible, temporarily change the Alarm Active Time to 24/7 to conduct the test. When you are done, return the Active Time setting to 10PM-7AM.
2. Review the test results to ensure accuracy and reliability.

3. Ongoing Monitoring and Maintenance

- **Routine Checks:**

1. Perform weekly checks to ensure configurations remain consistent.
2. Recalibrate settings if discrepancies are found.

- **Incident Response:**

1. If an undetected event occurs, review camera settings immediately.
2. Adjust settings as necessary and document changes.

4. Documentation and Reporting

- **Configuration Logs:**

1. Maintain detailed logs of all configurations and changes.
2. Document any issues encountered, and the resolutions applied.

- **Audit Reports:**

1. Conduct monthly audits of camera settings.
2. Generate and submit audit reports to the Technical Support Manager.

5. Continuous Improvement

- **Feedback Loop:**

1. Collect feedback from on-site teams regarding AI performance.
2. Adjust and update configurations based on feedback and new standards.

Troubleshooting Common Issues

1. Frame Rate Discrepancies:

- Check the network bandwidth and ensure it supports the configured frame rates.
- Verify the camera firmware is up to date.

2. Detection Failures:

- Ensure the camera lens is clean and unobstructed.



- Review recent changes to Ai algorithms or software updates.

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Approval

i3 Representative	Global Axiom Representative
Name:	Name:
Title:	Title:
Date:	Date:
Signature:	Signature: